

OE-EC704B — Optimization Technique

Semester VII • 3L:0T:0P • 3 Credits

Module 2 Exam Notes: Single-Variable Optimization Algorithms

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1. Optimality Criteria for Single-Variable Functions

Optimality criteria determine whether a candidate point x^* is a local minimum, local maximum, or an inflection point.

1.1 Necessary Condition (First-Order)

Theorem: Necessary Condition

If a function $f(x)$ is continuous and differentiable at x^* , and x^* is a local extremum (minimum or maximum), then the first derivative at x^* must vanish:

$$f'(x^*) = \left. \frac{df}{dx} \right|_{x=x^*} = 0 \quad (1)$$

Such a point x^* is called a **Stationary Point**.

1.2 Sufficient Condition (Second-Order)

Theorem: Sufficient Condition

Let $f(x)$ be twice differentiable at the stationary point x^* .

- If $f''(x^*) > 0$, then x^* is a **local minimum**.
- If $f''(x^*) < 0$, then x^* is a **local maximum**.
- If $f''(x^*) = 0$, the test is inconclusive. We must evaluate higher-order derivatives:

$$f'(x^*) = f''(x^*) = \dots = f^{(n-1)}(x^*) = 0, \quad f^{(n)}(x^*) \neq 0 \quad (2)$$

- If n is **odd**, x^* is an **inflection point** (neither minimum nor maximum).
- If n is **even** and $f^{(n)}(x^*) > 0$, x^* is a **local minimum**.
- If n is **even** and $f^{(n)}(x^*) < 0$, x^* is a **local maximum**.

2. Bracketing Methods

Bracketing methods find an initial interval of uncertainty containing the optimum before applying fast search routines.

2.1 Exhaustive Search Method

Exhaustive search evaluates a function at equally spaced points in an interval $[a, b]$ to find the sub-interval containing the minimum.

Exhaustive Search Algorithm

Let the initial interval be $[a, b]$, and let the number of evaluations be n .

1. Calculate the step size:

$$\Delta x = \frac{b - a}{n} \quad (3)$$

2. Define evaluation coordinates: $x_i = a + i \cdot \Delta x$, for $i = 0, 1, \dots, n$.
3. Evaluate $f(x_i)$ sequentially for $i = 1, 2, \dots, n - 1$.
4. Identify three consecutive points x_{k-1} , x_k , and x_{k+1} such that:

$$f(x_{k-1}) > f(x_k) \quad \text{and} \quad f(x_k) < f(x_{k+1}) \quad (4)$$

5. The minimum lies bracketed in the interval $[x_{k-1}, x_{k+1}]$. The new interval of uncertainty is:

$$L_1 = 2 \cdot \Delta x = \frac{2(b - a)}{n} \quad (5)$$

3. Region-Elimination Methods

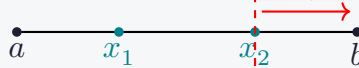
Region-elimination methods systematically reduce the interval of uncertainty by comparing objective values at test points.

3.1 Working Principle of Region Elimination

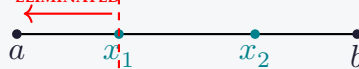
If we assume $f(x)$ is **unimodal** (has only one minimum in the interval $[a, b]$), we can eliminate a portion of the interval by evaluating two test points x_1 and x_2 (where $x_1 < x_2$):

Unimodal Region Elimination Logic

If $f(x_1) < f(x_2)$, minimum cannot lie in $(x_2, b]$



If $f(x_1) > f(x_2)$, minimum cannot lie in $[a, x_1)$



3.2 Interval Halving Method

This method eliminates exactly half of the interval at each stage using three evaluations (x_0 , x_1 , and x_2).

Interval Halving Algorithm

Let the current interval be $[a, b]$.

1. Calculate the midpoint: $x_0 = \frac{a+b}{2}$.
2. Calculate quarter points:

$$x_1 = a + \frac{b - a}{4}, \quad x_2 = b - \frac{b - a}{4} \quad (6)$$

3. Evaluate $f(x_1)$, $f(x_0)$, and $f(x_2)$.
4. Compare function values to eliminate region:
 - If $f(x_1) < f(x_0)$, the new interval is $[a, x_0]$.
 - If $f(x_2) < f(x_0)$, the new interval is $[x_0, b]$.
 - If $f(x_1) \geq f(x_0)$ and $f(x_2) \geq f(x_0)$, the new interval is $[x_1, x_2]$.
5. In all cases, the new interval width is exactly:

$$L_{new} = \frac{b - a}{2} \quad (7)$$

3.3 Fibonacci Search Method

Fibonacci search is the mathematically optimal region-elimination method. It uses the Fibonacci sequence $F_0 = 1$, $F_1 = 1$, $F_n = F_{n-1} + F_{n-2}$ to calculate placement ratios.

Fibonacci Search Equations

For a pre-selected total number of function evaluations n :

1. The initial two test points x_1 and x_2 in $[a, b]$ are placed at:

$$x_1 = a + \frac{F_{n-2}}{F_n}(b - a) \quad (8)$$

$$x_2 = b - \frac{F_{n-2}}{F_n}(b - a) = a + \frac{F_{n-1}}{F_n}(b - a) \quad (9)$$

2. After region elimination, the remaining sub-interval is evaluated. At the k -th step, the new test point is placed at:

$$x_k = \text{Remaining Limit} \pm \frac{F_{n-k-1}}{F_{n-k+1}}(\text{Current Interval Width}) \quad (10)$$

3. The final reduction ratio of the interval of uncertainty after n iterations is:

$$\frac{L_n}{L_1} = \frac{1}{F_n} \quad (11)$$

4. Point Estimation Methods

Unlike region elimination, point estimation methods fit a smooth polynomial through known points to estimate the local minimum directly.

4.1 Successive Quadratic Estimation (SQE)

SQE fits a quadratic function $q(x) = Ax^2 + Bx + C$ through three test points $x_1 < x_2 < x_3$ with known function values f_1, f_2, f_3 .

SQE Minimum Point Formula

The minimum x^* of the approximating quadratic polynomial is given by:

$$x^* = \frac{1}{2} \frac{(x_2^2 - x_3^2)f_1 + (x_3^2 - x_1^2)f_2 + (x_1^2 - x_2^2)f_3}{(x_2 - x_3)f_1 + (x_3 - x_1)f_2 + (x_1 - x_2)f_3} \quad (12)$$

Alternatively, we can express the calculation using coordinate differences:

$$x^* = x_2 + \frac{1}{2} \frac{(x_2 - x_1)^2(f_2 - f_3) - (x_2 - x_3)^2(f_2 - f_1)}{(x_2 - x_1)(f_2 - f_3) - (x_2 - x_3)(f_2 - f_1)} \quad (13)$$

If x^* is close to x_2 within a tolerance limit, x^* is taken as the optimum. Otherwise, the worst point among $\{x_1, x_2, x_3\}$ is replaced by x^* , and the calculation is repeated.

Quick Revision — Module 2

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Key Formulas & Equations

Necessary Condition	$f'(x^*) = 0$ (stationary point requirement).
Sufficient Condition	$f''(x^*) > 0 \Rightarrow$ local min; $f''(x^*) < 0 \Rightarrow$ local max.
Interval Halving Width	$L_{new} = \frac{1}{2}L_{old}$ (achieved with 3 evaluations).
Fibonacci Point 1	$x_1 = a + \frac{F_{n-2}}{F_n}(b - a)$.
Fibonacci Reduction Ratio	$\frac{L_n}{L_1} = \frac{1}{F_n}$ (where F_n is the n -th Fibonacci number).
SQE Minimum Point	$x^* = x_2 + \frac{1}{2} \frac{(x_2 - x_1)^2(f_2 - f_3) - (x_2 - x_3)^2(f_2 - f_1)}{(x_2 - x_1)(f_2 - f_3) - (x_2 - x_3)(f_2 - f_1)}$.

One-Line Definitions

- **Stationary Point:** A point at which the first derivative of a differentiable function is zero.
- **Inflection Point:** A point on a curve at which the concavity changes, where $f''(x) = 0$ and $f'''(x) \neq 0$.
- **Unimodal Function:** A function that has exactly one local extremum (minimum or maximum) in a given interval.
- **Bracketing Method:** A search algorithm designed to find an initial interval containing the optimum.
- **Region Elimination:** Systematically discarding a sub-interval of search space using unimodality comparisons.
- **Fibonacci Search:** An optimal region-elimination search algorithm utilizing Fibonacci ratios.
- **Successive Quadratic Estimation:** An interpolation algorithm that fits quadratic curves through test coordinates to locate the minimum.

Frequently Asked / Viva Questions

Q1. State the necessary and sufficient conditions for a local minimum of a single-variable function.

Necessary condition: $f'(x^*) = 0$. Sufficient condition: $f''(x^*) > 0$.

Q2. What happens if $f'(x^*) = f''(x^*) = 0$ at a candidate point?

The test is inconclusive. We must evaluate higher-order derivatives. If the first non-zero derivative is of odd order, it is an inflection point. If it is of even order and positive, it is a local minimum.

Frequently Asked / Viva Questions (contd.)

Q3. Define a unimodal function and state why it is important for region-elimination methods.

A function is unimodal if it has only one optimum in a given range. Unimodality guarantees that comparing function values at two points ($f(x_1)$ and $f(x_2)$) yields a valid sub-region to eliminate.

Q4. How does the Interval Halving method work?

It divides the interval into four equal parts using three points ($x_1 < x_0 < x_2$). Comparing their function values allows the elimination of half the interval ($L_{new} = 0.5L_{old}$).

Q5. Why is the Fibonacci search method considered optimal among region-elimination techniques?

For a pre-specified number of evaluations n , it achieves the maximum possible reduction in the interval of uncertainty ($1/F_n$) without requiring any derivative information.

Q6. What is the relation between Fibonacci search and Golden Section search?

Golden Section search is the limiting case of Fibonacci search as the number of evaluations $n \rightarrow \infty$. It uses the constant Golden Ratio $\gamma \approx 0.618$ instead of discrete Fibonacci fractions.

Q7. Write down the quadratic estimation formula for the minimum point x^* .

$$x^* = \frac{1}{2} \frac{(x_2^2 - x_3^2)f_1 + (x_3^2 - x_1^2)f_2 + (x_1^2 - x_2^2)f_3}{(x_2 - x_3)f_1 + (x_3 - x_1)f_2 + (x_1 - x_2)f_3} \text{ for three ordered points } x_1 < x_2 < x_3.$$

Q8. Differentiate between bracketing methods and point estimation methods.

Bracketing methods find an interval containing the optimum (e.g., Exhaustive Search). Point estimation methods fit functional approximations (e.g., quadratic curves) to predict the exact coordinate of the optimum.

Q9. Under what conditions does the successive quadratic estimation (SQE) method converge?

SQE converges rapidly if the objective function is smooth, continuous, and has a unimodal shape resembling a parabola in the neighborhood of the optimum.

Q10. What is the significance of the Fibonacci numbers sequence in search algorithms?

The Fibonacci numbers satisfy the property $F_{n-2}/F_n \rightarrow 0.382$ and $F_{n-1}/F_n \rightarrow 0.618$, which balances the spacing of evaluation points so that previous calculations are reused in subsequent steps.